FourSight Intent Classification — Technical Report

Seven-way first-hop intent router · single-turn model: fine-tuned **microsoft/deberta-v3-small** + auditable rule layer · contextual resolver for multi-turn chat · calibrated *needs_review* flag. All metrics are pooled out-of-fold results of a grouped, seeded 5-fold CV on the 1,904 training rows; the hidden test set was never used for any decision.

A. Problem, data findings and method

FourSight must route each question to one of seven backends; a wrong route means the right system is never queried, and OVERALL = 0.6·macro-F1 + 0.4·accuracy makes the three rare classes (chitchat 17, disruptions 28, other 55 rows) as important as realtime_query (782). EDA (*reports/eda.md*) drove four design decisions. (1) **Label noise is structural**: 165 exact-duplicate groups exist and 37 of them carry conflicting labels (e.g. a bare 10-digit id labelled both shipment_information and realtime_query; 'yes' labelled three ways), capping training-set accuracy at 98.0%. (2) **Validation must be grouped**: with plain stratified folds 319 of 1,904 validation rows (17%) have a verbatim twin in their training fold, so every number here uses StratifiedGroupKFold on normalised text. (3) The triad boundary is semantic, not lexical: reference numbers appear in 54% of realtime_query but also 21% of shipment_information rows; temporal-past and aggregate cues mark analytics (21% / 34% of its rows). (4) 15% of rows are ≤3 words and many are context fragments ('both', 'scheduled', 'eta'), which no single-turn model can resolve — this motivates the multi-turn design and the review flag.

Method. Baselines first: TF-IDF (word+char n-grams) with logistic regression / linear SVM, and frozen sentence embeddings (MiniLM, bge-small, multilingual-e5-small) with logistic regression; then fine-tuned compact encoders (DistilBERT, DeBERTa-v3-small, RoBERTa-base, multilingual-e5-small, bge-small) with a seeded fp32 training loop: AdamW, linear warm-up, weight decay, gradient clipping, √-inverse-frequency class weights, head+tail truncation to 128 tokens (pasted tables put the question last), early stopping on OVERALL. Temperature scaling, the review threshold and the rule layer were all fitted/evaluated on out-of-fold predictions (cross-fitted: fitted on four folds, applied to the fifth). The final artefact retrains the selected configuration on all 1,904 rows (3 epochs; median CV best epoch) after model selection was frozen.

Determinism. Inference is a pure function: fp32 on CPU, eval mode (no dropout), fixed thread count, *torch.use_deterministic_algorithms(True)*, fixed-size batches processed in input order, softmax with a fixed temperature, stable argmax (lowest label index on exact ties), and rules that are plain regexes. *tests/test_determinism.py* verifies bit-identical probabilities across repeated calls, fresh model loads and different batch compositions, plus valid output on empty, NaN, 20k-character, unicode and emoji inputs. Across different hardware/BLAS builds logits can differ at the 1e-6 level, which flips a label only on an exact near-tie; that is the stated guarantee, not universal bitwise equality.

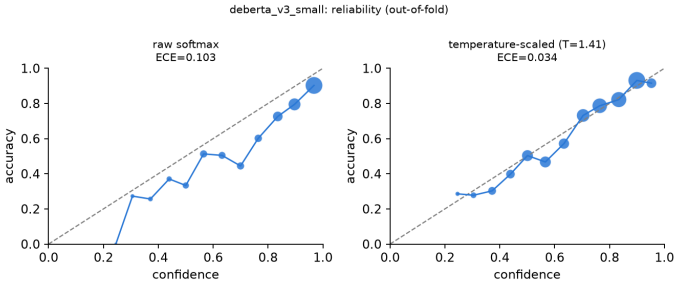
B. Validation results

model	acc	macro-F1	OVERALL	±fold	triad err	disr/chit/other F1
deberta_v3_small	0.707	0.611	0.649	0.031	380	0.75/0.80/0.19
bge_small_ft	0.727	0.581	0.640	0.026	353	0.61/0.67/0.09
e5_small_ft	0.717	0.569	0.628	0.029	367	0.76/0.75/0.00
roberta_base	0.717	0.568	0.628	0.034	361	0.78/0.33/0.16
distilbert	0.694	0.570	0.619	0.046	379	0.75/0.40/0.24
ensemble_top3 (deberta+bge+e5)	0.734	0.620	0.666	0.012	352	0.78/0.86/0.08
ensemble_top4 (deberta+bge+e5+roberta)	0.739	0.616	0.665	0.011	349	0.75/0.79/0.09
deberta_v3_small + rules (selected)	0.708	0.615	0.652	0.031	380	0.79/0.80/0.19

class	P	R	F1	n
shipment_information	0.32	0.25	0.28	256
realtime_query	0.72	0.83	0.77	782
analytics	0.68	0.52	0.59	257
disruptions	0.79	0.79	0.79	28
chitchat	0.78	0.82	0.80	17
knowledge_base	0.87	0.90	0.89	509
other	0.23	0.16	0.19	55

Selected: fine-tuned deberta_v3_small + rule layer — OOF accuracy 0.708, macro-F1 0.615, **OVERALL 0.652** (95% paired-bootstrap CI 0.626–0.676). Highest pooled OOF OVERALL among deployable single-model options under the pre-stated policy. It is within the 0.005 tie band of roberta_base + rules (0.6532), deberta_v3_small (0.6491); the tie-break prefers the stronger learned model without rules (deberta_v3_small 0.649 vs roberta_base 0.628, deberta_v3_small 0.649) and lower CPU latency (10.4 ms/row vs 11.1, 10.4). The rule layer adds +0.003 OVERALL (largest class deltas: disruptions +0.03, analytics -0.00, other +0.00) and no class loses more than 0.03 F1. The best ensemble (ensemble_top4 + rules) scores 0.666 (+0.013, 4/5 fold wins, min class ΔF1 -0.103) and fails the ensemble gate, so the single model ships. Frozen-embedding logistic regression reaches 0.629 and TF-IDF 0.580; fine-tuning buys most of its gain on knowledge_base and realtime_query precision, while the frozen baselines are competitive on the rare classes because class-balanced logistic regression is less prone to collapsing 17-example classes.

gold \ pred	shipme	realti	analyt	disrup	chitch	knowle	other
shipment_infor	64	157	15	0	1	10	9
realtime_query	86	647	29	3	2	11	4
analytics	24	69	135	1	0	23	5
disruptions	3	2	2	20	0	0	1
chitchat	0	0	1	0	14	2	0
knowledge_base	10	15	13	0	1	457	13
other	12	13	1	1	0	19	9



Where it fails. Confusion matrix above is the un-ruled deberta_v3_small OOF; 380 of 558 errors lie inside the operational triad, dominated by shipment_information→realtime_query (157) and realtime_query→shipment_information (86). Error categories: triad: semantic (283, mean conf 0.60); knowledge_base vs operational (113, mean conf 0.60); short/underspecified (84, mean conf 0.58); triad: temporal/aggregate cue (38, mean conf 0.57). Representative high-confidence misses: *'Origin Site Code Destination Site Code Material Material Description M'* (gold realtime_query, predicted analytics at 0.94); *'How many shipmets total last week'* (gold realtime_query, predicted analytics at 0.87); *'is there a way to see which set up we have with each carrier in fourki'* (gold analytics, predicted knowledge_base at 0.96); *'eta'* (gold shipment_information, predicted realtime_query at 0.89). Rows belonging to conflicting-label duplicate groups are classified correctly only 31% of the time and account for 10% of all errors; short (≤2-word) inputs reach 59% accuracy. These are precisely the inputs the multi-turn layer and the review flag are designed for.

C. Multi-turn architecture (implemented resolver + proposed contextual encoder)

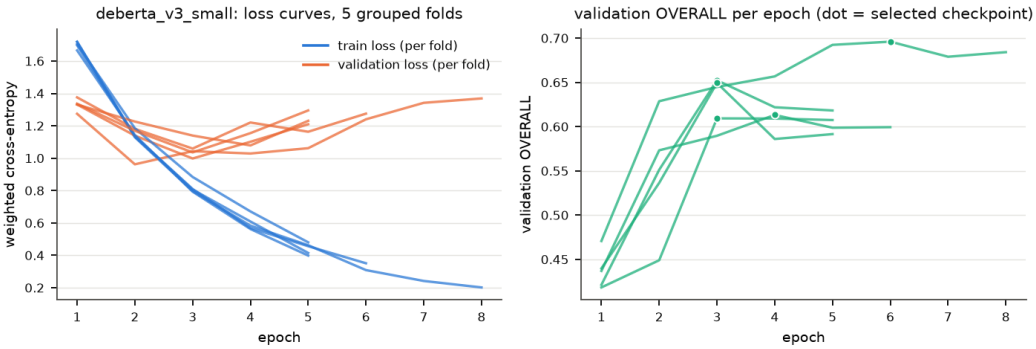
Implemented (*foursight_intent/conversation.py*, covered by *tests/test_conversation.py*): a **ConversationState** per session holds the last ≤6 user turns with their predicted intents, confidences and probability vectors, the *active reference ids* (shipment/PO/container numbers), *active topic* (last confidently resolved intent), *active time expression* ('last week'), and freshness counters (15-minute expiry, topic decay after 3 unrelated turns). Deterministic **follow-up detectors** flag elliptical turns ('what about last month?', 'and for Q2?'), pronoun references ('where is it now'), bare reference ids, bare time ranges ('last 7 days'), confirmations ('yes', 'both') and explicit topic switches / greetings. The **resolver** then applies a fixed precedence: a greeting or topic switch expires the context; a confident single-turn prediction ($p \geq 0.70$) always wins unless the turn is a pure confirmation or bare time range (so 'how do I set alerts for it?' after a tracking question is knowledge_base, never inherited realtime_query); a detected follow-up with fresh context blends $p = (1–w)\cdot p_{current} + w\cdot prior$, where prior is the decayed average of previous turn probabilities and w = 0.5 (0.7 for confirmations / bare time ranges), followed by bounded cue shifts inside the operational triad only (a past time range or aggregate word moves mass to analytics, a bare id to realtime_query, disruption vocabulary to disruptions). New ids in the turn override; otherwise follow-ups inherit the active ids and time range, which are returned to the backend together with the label. Low blended confidence raises *needs_review*; a context-only confirmation below 0.70 is always flagged. Every decision carries a human-readable *reason* and the single-turn label is logged beside the contextual one for audit.

Proposed production extension. Once real labelled conversations exist, fine-tune the encoder on a structured input `[CONTEXT] prev_intent=analytics ; prev_turn=<text> [CURRENT] <text>` (previous *user* turns only, most recent first, ~96 context + 128 current tokens), trained with (a) real sessions, (b) synthetic ellipsis rewrites of labelled single-turn questions and, crucially, (c) context-independent pairs so the model learns to ignore history instead of copying prev_intent. The deterministic resolver remains the baseline to beat and the fallback when the session store is down. **Evaluation.** No multi-turn accuracy is claimed: the dataset has no histories, and the eight scripted scenarios in *synthetic_scenarios()* are a functional smoke test, not a benchmark. The real benchmark: ~2,000 anonymised sessions (2–8 turns), every turn labelled *given the history* plus a 'context required?' bit and resolved entities, 20% double-annotated (target $\kappa \geq 0.8$); report OVERALL on all turns, on the context-required subset (what the resolver should improve) and the inheritance-error rate on the context-not-required subset (what blind inheritance would damage), with paired bootstrap over sessions; then shadow-mode and canary before full rollout.

D. Uncertainty, review flag and new intents

Temperature scaling (T = 1.41, fitted on OOF logits) reduces ECE from 0.103 to 0.034 (cross-fitted 0.024) and NLL from 0.904 to 0.843; mean calibrated confidence is 0.77 on correct and 0.59 on wrong predictions, and only 20 of 558 errors exceed 0.9. **needs_review** = calibrated max-probability < 0.538, the largest threshold flagging ≤20% of validation traffic. On validation it flags 20.0% of questions, catches 40.0% of all errors with precision 0.59 (cross-fitted: flag 21.3%, error recall 42.5%, precision 0.59); accuracy on the auto-answered 79% rises to 0.786 (OVERALL 0.698), AURC 0.146. The flag is metadata: every row still receives exactly one of the seven labels, and the threshold is a product dial (the full sweep is in *analysis_deberta_v3_small.json*). Confidence is a calibrated estimate, not a guarantee. On the 566 hidden-test questions the same threshold flags 26% (mean calibrated confidence 0.66 vs 0.72 out-of-fold): the test questions look somewhat harder or shifted (more French, more pasted templates), which is exactly what the flag is for; nothing was tuned on them.

A brand-new intent in production is a different problem from the existing *other* class (out-of-scope questions, answered by a fallback). Detection combines the calibrated confidence/margin, a rising share of *other*, an out-of-distribution score (cosine distance of the encoder embedding to its k nearest training neighbours — a monitoring signal, not a routing input) and downstream 'no result' / re-ask events. A weekly discovery loop clusters flagged questions (HDBSCAN on embeddings), a product owner triages clusters into noise / recall gap of an existing class / genuine new destination; the latter gets a definition, ≥100 annotated examples with agreement checks, and a **versioned taxonomy** (labels appended, ids never re-ordered; the artefact and router carry *taxonomy_version*). Retraining reuses the same seeded grouped-CV pipeline with acceptance gates (no existing class loses >1 F1 point on the frozen validation set; a golden boundary-case file must be predicted identically unless the change is approved), then shadow mode → 5% canary → full rollout with automatic rollback on review-rate or mis-route drift. The router fails closed: an unknown label maps to *other*. No eighth label is introduced into the current evaluation.



Loss curves and validation OVERALL per epoch for the five grouped folds of deberta_v3_small (dots = early-stopped checkpoints; the offline W&B runs hold the same curves per fold). Candidate comparison with bootstrap CIs: *reports/figures/candidate_comparison.png*.

E. Reproducibility

seed	42 (python, numpy, torch, shuffling generator; grouped 5-fold seed 42)
model	microsoft/deberta-v3-small → 7-way head, fp32; 3 epochs on all rows; temperature 1.406; review threshold 0.540; rule layer max_conf 0.6
hyper-parameters	lr 5e-05, batch 16, wd 0.01, warm-up 0.1, clip 1.0, class weight sqrt_inv, max_len 128 (+16 tail), patience 2
hardware	Apple M1 Max, 32 GB. CV fine-tuning on MPS (11 min / 5 folds); final fit on CPU, 8 threads (7 min). Inference on CPU: 12 ms/question batched (5 ms single). Any ≤24 GB GPU or a CPU suffices.
software	Python 3.11.15, torch 2.15.0, transformers 5.22.1, macOS-15.0.1-arm64-arm-64bit
inference	pip install -e . ; python scripts/generate_submission.py (writes + validates test_predictions.csv) ; python scripts/predict.py "<question>" ; pytho scripts/predict.py --conversation (multi-turn demo)
links	GitHub: NOT PUBLISHED (repository prepared locally; no public remote authorised in this session) · Hugging Face: NOT PUBLISHED (weights + model card in models/final/; upload via scripts/publish_artifacts.py --push) · W&B: NOT PUBLISHED (37 offline runs in wandb/ with loss curve per-class OVERALL and configs; `wandb sync wandb/offline-run-*` once credentials exist)